

Solution Requirements

Date:	29 September 2026
Team ID:	SWTID-2026-5955
Project Name:	PocketSmart AI - Your Smart Budget & Recommendation Assistant
Maximum Marks:	4 Marks

Define Functional and Non-Functional Requirements

- **Functional Requirements:** Define what the system should do; the specific behaviors, features, and capabilities that the application must provide to meet user needs.
- **Non-Functional Requirements:** Define how the system performs; qualities such as performance, reliability, scalability, security, and usability that ensure the system operates effectively.

Step 1: Functional Requirements

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely register/login via email/ password, JWT token issued at /token endpoint, session management via /session-info. Validates credentials, hashes passwords with <i>bccrypt</i> .	High
2	Authorization levels	Roles - End User can access Home/Party/ Jewelry planners and view history. Admin can manage users, monitor recommendations, view logs.	High
3	External interfaces	Gemini 1.5 Flash Pro API for AI recommendations, Third-party E-commerce APIS Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO for product sourcing, Flask/FastAPI backend with frontend HTML/CSS/JS.	High

Functional Requirements (Continued)

S.No	Functional Area	Requirement Description	Priority
4	Transactions processing	System processes budget inputs and generates recommendations via /generate-home, /generate-party, /generate-jewelry routes. Handles data exchange between UI and Gemini AI, validates budget constraints, ensures transactional integrity.	High
5	Reporting	Dashboard to display AI-generated recommendations with price, platform links. History page /history retrieves past queries. Export recommendations as PDF/JSON, displays budget breakdown charts.	Medium
6	Business Rules & Validation	Budget must be > 0; room quantity validation; guest count > 0; outfit image must be JPG/PNG <5MB. AI must allocate budget proportionally without exceeding total. Fallback to default recommendations if AI fails.	High
7	Compliance to Laws or Regulations	Complies with GDPR for user data privacy, secure password storage, data encryption, no sharing of personal images without consent. Follows e-commerce affiliate compliance.	Medium
8	Other	Image upload for jewelry matching; multi-category planners; session-data personalization; CORS enabled for secure frontend-backend communication.	Medium

Step 2: Non-Functional Requirements

S.No	NFR Category	Requirement Description	Target Metric
1	Performance	System response time for recommendation generation under normal load	≤ 3 seconds for 95% of requests
2	Scalability	System shall handle concurrent users and scale to increasing request volume	Support 500+ concurrent users; auto-scale as load increases
3	Security	Data encrypted in transit and at rest; secure authentication and authorization	TLS 1.3 encryption; passwords hashed via bcrypt (SHA-256+ salt)
4	Reliability / Availability	System should be highly available with graceful failure and fallback mechanisms	99.5% uptime; fallback to cached/default recommendations if AI unavailable
5	Usability	User interface shall be intuitive, responsive, and accessible across devices	Responsive design for desktop/mobile; WCAG 2.1 AA compliance

3.2 Non-Functional Requirements (Continued)

#	Requirement Area	Description	Target
5	Usability & Accessibility	Intuitive responsive UI built with HTML/CSS/JS Jinja2, mobile-friendly, clear forms for budget input, accessible color contrast, WCAG 2.1 AA compliance.	Responsive design for desktop/mobile; WCAG 2.1 AA, user can generate recommendation in <3 clicks.
6	Other	Maintainability - modular Flask/FastAPI code with routes/services/models separation, Portability - deployable on IBM Cloud, local, Docker, Compatibility with modern browsers Chrome/Firefox/Safari.	Code modularity >80%, Docker support, 99% browser compatibility.

Note:

PocketSmart AI integrates Gemini 1.5 Flash Pro with e-commerce APIs to deliver budget-smart recommendations.